SENIOR CAPSTONE PROJECT PROPOSAL

AutoEat

A low-cost, learning-based robotic cell for autonomous meal preparation

A proposal to design and build a self-contained robot that plans, orders, and physically prepares a nutritionally complete meal with no human in the loop — scoped honestly to a single, safe, buildable first dish, and grounded in an independent feasibility assessment. We are seeking a faculty advisor and access to departmental lab resources to carry this through a twelve-month capstone.

PREPARED FOR	Prospective Faculty Advisor — Robotics / Machine Learning
PREPARED BY	Rami Abukhaters (proposing capstone lead)
SCOPE	Layer 3 physical prep robot (v1), integrated with software planning & procurement
DURATION	12 months, part-time student team
EST. HARDWARE COST	~$400–750 (single build) with makerspace printing
REQUESTED OF ADVISOR	Technical advising + makerspace, lab, and GPU/cluster access
DATE	August 2026

1 Executive summary

AutoEat proposes to remove food from a person's daily decision-and-labor budget by automating three distinct problems: **planning** a nutritionally sound menu, **ordering** the ingredients, and **physically preparing** the meal. The first two are software problems we can solve this year. The third — physical preparation — is the research frontier, and the heart of this capstone.

Rather than chase a general-purpose kitchen robot (which is beyond the state of the art for anyone), we scope v1 to a single, deliberately constrained task: a fixed-base, low-cost robotic arm that assembles **one cold dish — overnight oats in a pre-portioned bowl** — by manipulating a spoon and light ingredient cups, with the manipulation policy learned from human demonstration via imitation learning on the open-source LeRobot stack. No heat, no blades, nothing over ~400 g moves under robot power, so the worst realistic failure is a dropped ingredient, not a fire or a cut.

This scope is not a guess. An independent feasibility assessment of the project (attached in summary in Section 8) rated v1 **“Feasible, with caveats”** at moderate-to-high confidence, judged the technical risk to be largely *engineering* risk rather than open research risk, and confirmed that comparable single-task manipulation results are already routine in the low-cost robotics community. The parts budget is real (~$120–340 in servos and cameras, plus contingency), the timeline is achievable if the team treats data collection and iteration — not assembly — as the main event, and the steepest challenge is the empirical reliability loop, which is exactly where a faculty advisor materially de-risks the project.

> **WHAT WE ARE ASKING OF YOU**
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> We are seeking (1) a faculty advisor to guide the manipulation-reliability and evaluation work that is the technical core of the project, and (2) access to departmental resources that turn this from a “maybe” into a “yes”: makerspace 3D printing for the arm, a bench and lab space for the prep cell, and GPU or cluster time for policy training. In return you sponsor a project with a crisp, demonstrable end state and a clean, well-documented research artifact. Details in Section 7.

2 Motivation & problem statement

Eating well every day is a surprisingly heavy cognitive and logistical load: deciding what to eat against nutrition goals, keeping ingredients stocked, and doing the repetitive preparation. Most existing “food automation” either solves only the digital slice (meal-planning apps, grocery delivery) or aims at a general-purpose humanoid chef that does not yet exist. There is a gap in the middle: a **narrow, reliable, end-to-end pipeline** that closes the loop from goal to prepared food for a constrained menu.

AutoEat targets that gap. The intellectual value for a capstone is that it forces a genuine, integrated robotics-and-ML system — perception, learned manipulation, safety, and software orchestration — onto a small budget and a realistic student timeline, and it produces a measurable result: a success rate on a physical task, driven up through disciplined iteration. It is the kind of contained, reliability-focused problem that teaches the parts of robotics that coursework usually cannot.

3 Project vision and honest scoping

The full AutoEat vision is a pipeline of three layers coordinated by an orchestrator. The core discipline of this proposal is knowing which parts are buildable now and which must be cut or deferred.

Layer	What it does	Difficulty	v1 decision
1 — Planner	Turns a person's profile and goals into a weekly menu, per-meal macro targets, and a pantry-aware shopping list.	Low (pure software)	**Build first.** LLM for meal variety, a nutrition database + solver for the numbers.
2 — Procurement	Resolves the shopping list to real products and places an order.	Low–medium *only if* the store robot is dropped	**Automate the order** through a delivery/pickup channel. No robot in a physical store.
3 — Prep cell	Physically assembles the dish in a purpose-built, environment-constrained station.	High — the frontier	**One constrained cold dish**, learned from demonstration. This is the research core.

> **THE SINGLE MOST IMPORTANT FRAMING**
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> Every hard part that sounds most exciting — the store-shopping robot, the general-purpose kitchen chef, lifting hot pots and pans — is exactly the part to cut in v1. Those are what would sink the project. What remains after cutting them is still a startlingly large slice of the dream, and it is genuinely buildable in a year.

The one decision that sizes everything: payload

Cost, safety, and difficulty all flow from how heavy the objects are that the arm must lift. A wooden spoon (50–100 g) and a light ingredient cup or bowl (150–400 g) are well within a cheap arm's envelope. A pot or pan — especially a full one at 2–6+ kg — is not, by a factor of four to ten, and would throw the project out of the hobby-arm tier and into an $8,000+ cobot with an entirely different safety class. We therefore freeze a rule into the v1 design:

> **THE V1 DESIGN RULE**
>
> **Nothing heavier than ~400 g ever moves under robot power.** The robot manipulates a spoon and light pre-portioned cups only. This single rule sizes the arm, the budget, and the safety envelope all at once — and it is exactly how commercial food robots dodge this problem.

4 Technical approach

4.1 System architecture

Three functional layers sit on two cross-cutting subsystems — **learning** and **safety** — coordinated by an orchestrator that holds system state, decides when to order and when to prep, triggers each layer, and logs everything for debugging and for the learning loop.

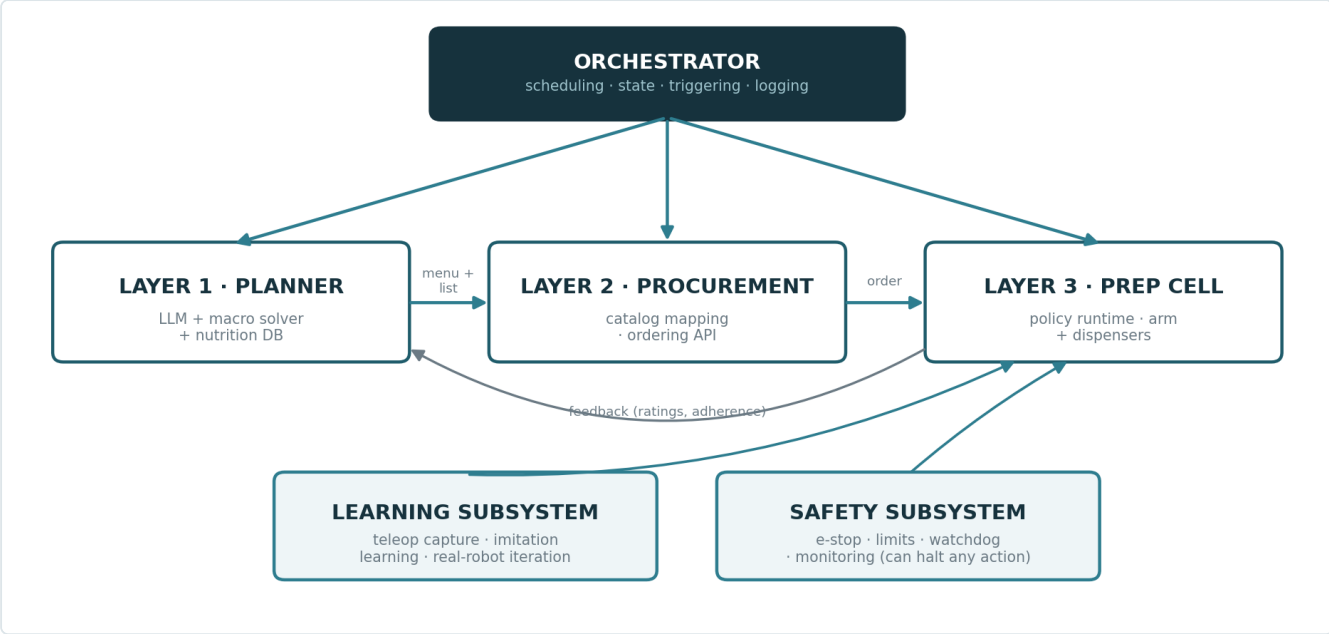


Figure 1. The AutoEat pipeline. The planner emits a machine-readable plan that procurement and the prep cell consume; the learning and safety subsystems cut across all three, with safety able to veto or halt any action.

4.2 Hardware base

The manipulator is the **LeRobot SO-101**: a 6-DOF, ~500 mm-reach, ~500 g-payload open-source arm whose structural parts are 3D-printed and whose only real cost is six Feetech STS3215 smart servos (~$13–15 each). Critically, those servos give position feedback, which imitation learning requires — this is the one component not to cheap out on. Perception uses two ordinary RGB USB webcams (one fixed overhead, one at the wrist), the standard, well-supported LeRobot configuration; a depth camera is treated as optional and is not required at this scale. A cheap MLX90614 IR thermometer is included only as an independent safety monitor, not for control, and is not critical for a cold v1.

4.3 Learning approach

The manipulation policy is trained by **imitation learning** (ACT, via the Hugging Face LeRobot stack) on human teleoperation demonstrations. A leader+follower arm pair lets an operator drive the leader by hand while the follower records synchronized observations and actions; a single-arm kinesthetic (hand-guided) fallback exists but produces noisier data, so we recommend the leader+follower pair for a contact-rich food task. Training runs on free or low-cost GPU (Google Colab or a university cluster), with datasets and

checkpoints versioned to ~400 GB of Google Drive that mounts straight into Colab. Simulation is used only to *augment* data, never as the primary trainer — for contact-rich manipulation, real demonstrations reliably beat pure sim-to-real.

4.4 Safety subsystem

Safety is cross-cutting and authoritative: it can veto or halt any action. It comprises a hardware e-stop and mechanical workspace limits, a watchdog that halts on anomaly or loss of camera perception, and a deliberate design bias toward benign failure modes. Because v1 has no blades and no open flame and lifts nothing heavy, the worst realistic outcome is a spill or a dropped ingredient. Food-contact parts are food-grade and cleanable rather than the porous printed plastic, and v1 is operated **attended only** — unattended operation is explicitly a later-version concern.

5 Feasibility — summary of the independent assessment

Before committing, the project was subjected to an independent technical, financial, and timeline review. Its overall verdict was **“Feasible, with caveats”** at moderate-to-high confidence for v1 as scoped. The dimension-by-dimension findings:

Technical	**Feasible, with caveats.** Rigid pick-and-place sub-steps reach 80–90% after tuning; granular/viscous scooping and pouring is lower and noisier (50–70% per step early). The honest hardest sub-problem is reliable spoon-to-cup scooping of granular/ viscous food from a fixed-camera policy — an engineering risk with a research-risk tail, not an unsolved problem.
Financial	**Feasible on parts.** The ~$120–340 servo-and-camera figure is real; true expected spend with contingency (filament, spare servos, incidentals, shipping) is roughly 1.5–2× that — still a very cheap capstone.
Timeline	**Feasible, front-loaded risk.** Twelve months is enough only if the team treats data collection and the train–evaluate–fix loop as the main work. The slump hits months 6–9 when the reliability plateau appears.
Team / skills	**Plausible for a 4th-year team.** No required skill (3D printing, servo bring-up, Python/ ML, data discipline) is out of reach. The steepest curve is the empirical debugging loop — diagnosing *why* a policy fails — which faculty guidance directly addresses.
Safety	**Adequate for v1.** The no-heat / no-blades / nothing-over-400 g envelope is the right posture. Remaining items (pinch points, food hygiene, attended-only operation) are standard and manageable.

THE SINGLE BIGGEST RISK — AND HOW WE BLUNT IT

Reliability collapse from step-chaining. End-to-end success is the *product* of per-step success rates, so a four-step chain at 70–80% per step yields only ~30–40% end-to-end even though every piece “works.” We reduce it four ways: pick the dish to minimize step count and avoid precise pours; consider per-step sub-policies with scripted transitions so failures isolate and debug cleanly; over-collect demonstrations on the worst step specifically; and define “success” concretely and early (e.g. “edible bowl assembled within tolerance in N of 10 trials”) so the team optimizes the right number.

6 Twelve-month work plan

The plan is deliberately front-loaded on getting the toolchain and data pipeline working, because the reliability grind — not assembly — is what consumes the year.

> **Months 0–1 — Foundations** `M1: parts ordered`
> Build the Layer 1 planner MVP (menu + macros + shopping-list output). Freeze the target dish and design the prep-cell layout on paper. Order hardware and confirm makerspace printing. Stand up procurement against a delivery/pickup channel in test mode.

> **Months 1–3 — Data rig and procurement live** `M2: teleop capturing data`
> Assemble the teleoperation rig (arm + fixed stations + two cameras). Begin collecting human demonstrations of the dish. Make procurement place real orders end-to-end from the planner's shopping list. Prove a trivial rigid pick-and-place (grab the spoon, move it) to ~80% first — this validates the entire toolchain before touching food.

> **Months 3–6 — First policy** `M3: first trained policy on real rig`
> Train an imitation-learning policy on the collected demonstrations. Stand up a simulation of the cell for data augmentation. Build the safety subsystem: e-stop, mechanical limits, watchdog, monitoring.

> **Months 6–9 — Close the loop** `M4: three layers integrated`
> Iterate the policy on the real rig — every failure shows where training or sim lied, so fix and retrain. Integrate the three layers so a menu becomes an order becomes a prepped dish with no human in the middle. Begin measuring reliability over N attempts. *This is the hardest window; protect it.*

> **Months 9–12 — Harden and demo** `M5: end-to-end demonstration`
> Push reliability up, validate safety under adversarial conditions, and produce the end-to-end demonstration: the system plans a day's meals, orders the ingredients, and physically prepares the target dish, repeatably, in front of an audience.

Success criteria & deliverables

- **Primary metric:** target-dish assembled within tolerance in at least **N of 10 consecutive attended trials** (target set with the advisor at project start; a reasonable stretch goal is 8/10).
- **Working prep cell:** the physical SO-101 rig, gripper fingertips, fixed stations, and two-camera perception setup.
- **Software pipeline:** planner → procurement → orchestrator, emitting machine-readable plans and placing real test orders.
- **Research artifact:** the demonstration dataset, trained policy checkpoints, an evaluation write-up of reliability vs. demonstration count, and reproducible documentation.
- **Final capstone deliverables:** a live end-to-end demo, a written report, and a recorded video of the full pipeline.

7 What we are requesting from you

Two things, and the feasibility review named both as the highest-leverage forms of support for a project at exactly this scope.

REQUEST 1 — FACULTY ADVISING

Guide the reliability and evaluation work that is the technical core

The assessment was explicit that the steepest curve is not any single technology but the **empirical debugging loop** — diagnosing why a policy fails and knowing whether the fix is more data, better data, camera stability, or task redesign. This is precisely where an advisor's experience compresses months of flailing into weeks. Concretely we would ask for periodic technical check-ins, guidance on experiment design and the evaluation protocol, and a sanity check on scope at each milestone.

REQUEST 2 — ACCESS TO DEPARTMENTAL RESOURCES

The resources that move the true project cost toward the price of a few webcams

Makerspace / lab 3D printing for the SO-101's large multi-part frame — the single biggest cost lever, and it takes the arm cost down to just the servos. **Bench and lab space** for the fixed prep cell, since the camera and scene must stay rigid across data collection and deployment. **GPU or cluster time** for policy training, which is safer and faster than relying on the free Colab tier alone across many iterations. Access to any existing lab arms or printers would further de-risk the build.

TWO HONEST GATING CONDITIONS

The review flagged that these resource questions are what would flip the project from “go” to “no-go”: if makerspace printing and GPU access are *unavailable*, true cost and complexity rise sharply. Neither is a technology failure — both are resourcing questions we would want to confirm with you early, before the team commits the year. That is a large part of why we are approaching you at the proposal stage rather than after starting.

8 Budget

The following reflects the minimal student build with makerspace printing, plus the contingency the feasibility review recommended adding (the headline parts figure omits filament, spare servos, and incidentals). Compute is budgeted at $0 assuming Colab free tier or cluster access.

Item	Notes	Cost (USD)
Follower-arm servos (6× STS3215)	Position-feedback smart servos — the one part not to cheap out on	$80–110
Leader arm for teleop (6× STS3215)	Recommended for a contact-rich food task; enables clean demonstrations	$90–110
Servo driver board + cables		$10–20
3D-printed arm parts	Free at makerspace; otherwise filament	$0–30
Two USB webcams	Overhead + wrist (standard LeRobot setup)	$30–50
Gripper fingertips (spoon + cup), stations, frame	Printed or scrounged	$0–40
MLX90614 IR sensor + microcontroller	Safety monitoring only (optional for cold v1)	$30–60
Compute (Colab free / university cluster)		$0
Contingency (spares, filament, incidentals, shipping)	Per feasibility review; ~1.5–2× headline parts	$120–250
Estimated total (leader + follower build, with contingency)		**~$550–750**

A single-arm kinesthetic build lowers the floor to roughly $400–550 but produces weaker demonstration data; we recommend the leader+follower pair as the better value for a manipulation task. All prices are approximate, US, and current as of August 2026; hardware prices move, so they will be re-checked before purchase.

9 Responsible scope

We want to name, briefly, that automating food preparation raises real questions about labor displacement and the homogenization of food. Our answer at the capstone scale is clean: **building this as a personal tool displaces no one** — it automates the operator's own labor, the same moral category as a dishwasher or bread machine. The harder questions live in a hypothetical commercial deployment, which is explicitly out of scope for this project. If the work ever moved in that direction, our stated design commitments would be to optimize for variety, nutrition, and quality rather than cost-per-meal, to target genuine unmet need (people who currently cannot cook for themselves) over pure labor-replacement, and to be honest about trade-offs. We

raise it here not as a disclaimer but because we think a serious project should be able to state where its ethical lines are.

One institutional note to settle early: if departmental printers, lab hardware, GPU clusters, or faculty inventive contribution are used, university IP policy may attach to project outputs. This is routine for capstones and usually easily resolved, but we would want to clarify ownership in writing with the department at the outset rather than discover it late. The open-source (Apache-2.0-style) licensing of the LeRobot/SO-ARM stack should simplify this.

10 Why this is a strong capstone

It is **contained but real**: a single, well-defined task that nonetheless exercises the full robotics-and-ML stack — perception, learned manipulation, safety, and system integration. It is **measurable**: success is a reliability number the team drives up through disciplined experimentation, which makes progress legible and the final result concrete. It is **cheap and de-risked**: the parts cost less than many textbooks, the method is a known-solved pattern in the low-cost manipulation community rather than an open research gamble, and an independent review has already pressure-tested the scope. And it is **demonstrable**: the end state is a machine that plans a meal, orders it, and makes it — a result an audience immediately understands. For a robotics or ML advisor, it offers a clean vehicle for teaching the empirical reliability engineering that coursework rarely reaches, and a well-documented dataset and evaluation artifact at the end.

> **RECOMMENDED FIRST TARGET DISH**
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> **Overnight oats in a single pre-portioned bowl.** Every hard edge is filed off: no heat, no cutting, forgiving of imprecise scoops, few steps (drop oats, add yogurt/milk cup, add toppings, stir), and the failure mode is a slightly messy but still-edible bowl rather than anything dangerous. An assembled yogurt-grain bowl is the near-equivalent fallback. Anything requiring precise liquid pouring or level measuring is deferred — earned later, not attempted first.

11 References

1. TheRobotStudio, *SO-ARM100 / SO-101 official repository and specifications* (6-DOF SO-101, STS3215 servos, open-source). github.com/TheRobotStudio/SO-ARM100
2. Hugging Face, *LeRobot SO-101 documentation* (follower uses 6× STS3215; assembly and setup). huggingface.co/docs/lerobot/so101
3. Feetech, *STS3215 servo specification* (~30 kg·cm stall torque @12V; 12-bit magnetic encoder; position/load/temperature feedback). Waveshare SO-ARM100/101 wiki; Feetech listing.
4. Hugging Face, *ACT policy documentation* (~50 demonstrations often sufficient for a single-variation task; ~100k training steps; a few GPU-hours). huggingface.co/docs/lerobot — ACT.
5. A. Karkada, *ACT real-world success rate on SO-101* (~70% on a 50-episode pick-and-place task), Medium, 2025; comparable LeRobot ACT tasks report 70–90% with careful data (arXiv 2409.17435; 2503.01543).
6. SO-101 build time and cost references: assembly ~2–4 hours; DIY build from ~$150, kits from ~$199 (roboticscenter.ai/hardware/so-101; vendor listings, 2026).

Feasibility ratings and risk analysis summarized in Sections 5 and 8 are drawn from the project's independent feasibility assessment (AutoEat — Layer 3 physical prep robot v1). Institution-specific items — makerspace printer availability and free-tier

GPU quota for this team — were flagged in that review as unverified and are among the resource questions this proposal asks to confirm.